Project Storm — Valuation and Bid Recommendation

Acquisition of a 50MW operational windfarm · valuation date 31 December 2012

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> **Note on data.** Project Storm, the vendor, the investor and all associated figures are illustrative and constructed for this case study. This is not a live mandate and does not represent any real project, fund or transaction.

1. Summary

Analysis	Result
Bid price for 100% of the equity	**£64.5m** (implied EV £61.0m, £1.22m/MW)
Downside: P90 output for ten years, then P75	Equity IRR falls from 12.0% to **6.9%**
Exit: sale to an infrastructure fund after two years at an 11% price	**13.2%** IRR
Power purchase agreement	Value-neutral at a **~4%** discount to the forward curve
Twelve-year amortising LIBOR swap	**1.64%** (4.64% all-in)
30% interest deductibility cap	**Nil** impact on price
Five-year subordinated loan at 9%	Equity IRR **13.9%** (+195bps)

Recommendation. Bid £64.5m for the equity — treat it as a ceiling and open nearer £60m. The vendor cannot raise project finance and must close by year end; the price is already £7.4m below the £71.9m book equity in its accounts.

Settle the financing before settling the price. Senior debt is capped at £51.2m by the 1.30x DSCR test, not by the 85% loan-to-value test, which would allow £57.4m. Moving the DSCR floor to 1.15x is worth £1.9m of price; moving the hurdle rate by 400bps is worth only £2.3m. The debt does the work.

Do not go materially above £64.5m without a PPA or a lower DSCR floor. At 84% gearing the equity is thin, and a 10% fall in the power price is a payment default, not merely a lower return.

2. The asset and the transaction

A 50MW windfarm, operating since January 2008, with 15 years of residual economic life. P75 output is 2,500 hours (125 GWh a year); P90 is 2,350 hours. Revenue is merchant power plus a regulated subsidy of £40/MWh (2012 prices, escalating at CPI less 1.5%) for 15 years from first operation — so the subsidy ends after 2022.

The vendor is balance-sheet constrained, has been unable to raise project finance, must close by 31 December 2012 and keeps the economic benefit of the asset to that date. The investor targets a 12% buy-and-hold equity IRR, with the option to sell down to a lower-return buyer after a few years.

Senior debt terms: 12-year tenor, LIBOR + 300bps, 200bps upfront fee, sized at the lower of 85% of fixed assets and a 1.30x DSCR on P75 cash flow, with 85% of principal hedged. Tax at 25%, with interest deductibility capped at 30% of EBITDA (excess carried forward). Transaction costs £0.5m.

3. Bid price

The power price

Supply must clear demand, so the marginal plant sets the price. The ranking is not static: coal costs escalate while CCGT costs fall, and the two cross in 2020, at which point renewables plus CCGT fall just short of demand and coal sets the price for two years. That gives a visible hump before coal retires and CCGT takes over again. A static merit order would miss it.

Year	Marginal plant	Real (£/MWh)	CPI index	Nominal (£/MWh)	Subsidy (£/MWh)
2013	CCGT	54.73	1.0250	56.09	40.40
2019	CCGT	47.90	1.1657	55.83	42.04
2020	Coal	49.68	1.1890	59.07	42.25
2021	Coal	50.67	1.2128	61.45	42.46
2022	CCGT	41.97	1.2370	51.92	42.68
2023	CCGT	39.78	1.2617	50.20	ends
2027	CCGT	30.64	1.3658	41.84	—

Full annual series: model, Calcs rows 23–50.

Debt capacity

Two tests; the lower governs. Sized on P75 and excluding working capital. Tax depends on interest and interest depends on the debt quantum, so the two are solved together — the convergence check in Calcs D118 reads nil.

Test	Capacity	Status
1.30x DSCR on P75 CFADS over 12 years	£51.2m	**Binds**
85% of fixed assets	£57.4m	£6.1m of headroom

Bridge to the price

Dividends do not depend on what is paid — the price only affects the day-one cheque — so the bid is a direct calculation, not a goal-seek.

£'000	
PV of dividends 2013–27 at 12% (= equity cheque)	14,800
Add: senior debt drawn	51,242
Less: transaction costs	(500)
Less: upfront debt fee at 2.00%	(1,025)
Equity purchase price payable to vendor	**64,518**
Less: cash acquired	(3,500)
Implied enterprise value	**61,018**

Equity IRR 12.0% · money multiple 1.81x · EV/2013 EBITDA 6.9x · gearing (debt/EV) 84% · unlevered project IRR 5.8% · equity 24% of price.

4. Downside — P90 for ten years, then P75

Equity IRR falls from 12.0% to 6.9%; the multiple from 1.81x to 1.46x. A 6% volume loss removes 42% of the equity return, because debt service is fixed and takes the first £6.4m of cash — the whole shortfall lands on the residual claim.

Year	EBITDA base	EBITDA P90	DSCR	Dividend
2013	8,851	8,166	1.23x	4,964
2016	8,897	8,197	1.22x	1,382
2019	8,529	7,839	1.21x	1,301
2022	7,863	7,201	1.21x	1,179

£’000. 2013 includes the £3.5m opening cash released.

Minimum cover is 1.21x, in 2022 — no payment default. But cover sits below the 1.30x sizing case in every year, so a distribution lock-up set at or above 1.20x would be within a whisker of tripping, and a lock-up traps cash rather than merely reducing it. Worth negotiating the lock-up at 1.15x and confirming covenant tests on P75 rather than actual production.

5. Exit after two years

An infrastructure fund buys the equity at 31 December 2014 on the same base case, assumes the existing debt and prices to an 11% levered return. Its price is the PV of the 2015–27 dividends at that rate.

£’000	
PV of dividends 2015–27 at 11%	11,062
Add: senior debt outstanding	42,335
Enterprise value at exit	53,397
Exit fee at 0.25% of EV	(133)
Net equity proceeds	**10,929**

Investor cash flows	2012	2013	2014
Total	(14,800)	5,408	12,842

A two-year IRR of 13.2% against 12.0% held to term — +120bps, on a 1.23x multiple. The exit equity value of £11.1m is below the £14.8m invested; that is not value destruction but a depleting 15-year annuity from which £7.3m has already been drawn. Most of the return comes from dividend yield and deleveraging, not multiple expansion.

6. Financial impact of a PPA

A PPA would replace merchant electricity revenue — about 58% of the top line to 2022 and all of it thereafter — with a contracted price, and the financial effect runs through the financing rather than the revenue line. Any PPA is struck at a discount to the forward curve, because the offtaker is paid to absorb price and volume risk, and that discount reduces EBITDA directly. Against it, contracted revenue supports materially more debt: lenders size merchant wind at around 1.30x but will lend against a strong PPA at 1.15x to 1.20x. Since the DSCR test is precisely what binds here, relaxing it converts straight into purchasing power. The extra debt is worth about £1.9m of price if the PPA is struck at the curve, while each 5% of discount costs roughly £2.5m — a break-even discount of about 4%.

Benefits. More and cheaper debt (a lower DSCR floor, a tighter margin, a longer tenor). It removes the dominant risk: merchant exposure is what takes cover to 1.21x under P90 and to 0.89x on a 10% price fall. A PPA running past 2022 is the only structural answer to the subsidy cliff, when EBITDA falls by three-quarters, and contracted assets clear at lower returns with infrastructure buyers, widening the exit.

Downsides. The discount is permanent value given away, and beyond ~4% it exceeds the financing gain. It forgoes upside — coal sets the price at £61/MWh in 2020–21, and a 10% price rise is worth 570bps of equity IRR on a merchant basis. And it brings counterparty, term and shape risk: a PPA shorter than the debt leaves a merchant tail, and a fixed-volume structure means a P90 year forces purchases in the market to cover the shortfall.

Discount to curve	DSCR floor 1.30x	1.20x	1.15x
0% (at the curve)	64,518	65,926	66,438
5%	62,014	63,362	64,128
10%	59,510	60,798	61,530
15%	57,006	58,234	58,932

Bid price at an unchanged 12% equity IRR, £’000.

7. Interest rate hedge

Hedging 85% of a reducing balance is an amortising swap, so the fixed rate is the notional-weighted average of the LIBOR forwards discounted at LIBOR — not a simple average of the curve:

Swap rate = $\Sigma\ (N_t \times F_t \times DF_t) \div \Sigma\ (N_t \times DF_t)$

Twelve-year amortising swap rate: 1.64%; all-in fixed cost 4.64% including the 300bps margin. That is well below the 2.08% simple average of the curve, because the notional is largest in the early years when LIBOR is lowest; quoting the simple average would overstate the hedged cost by ~44bps.

A swap at these levels carries real break costs. On a two-year exit it must be novated or unwound, and in a falling-rate environment that is a cash cost to the investor. Size the unwind risk before committing to the full 85%.

8. Interest deductibility limitation

Impact on the purchase price is nil; the 30% cap does not bind in any year. This is structural: the DSCR test caps senior debt at £51.2m, and financing costs peak at £2.07m in 2013 against a £2.66m cap. The tightest point is 2023, where headroom is still £0.39m — reflecting the subsidy expiring, not leverage. Senior debt would have to reach roughly £65.9m, or 98% of fixed assets, before the cap bit.

Year	EBITDA	30% cap	Financing cost	Headroom
2013	8,851	2,655	2,065	590
2021	9,115	2,735	867	1,868
2023	2,224	667	279	388

It becomes live with higher-cost junior capital (the subordinated loan lifts 2013 financing costs to £2.63m against a £2.66m cap) and in an EBITDA shock, since the cap tightens exactly when trading is weak.

Cap (% of EBITDA)	Senior debt	Bid price	Change
No limit / 30%	51,242	64,518	—
20%	51,222	64,485	(32)
15%	50,856	64,032	(486)
10%	50,190	63,236	(1,281)
Nil deduction	48,859	61,646	(2,872)

9. Subordinated loan

Five annuity payments at 9%, sized so post-senior cash covers the annuity 1.20x. The binding year is 2017, when post-senior cash is lowest (£1,921k), giving a maximum annuity of £1,601k and, at an annuity factor of 3.8897, a loan of **£6,227k**.

At the same £64.5m price	Base	With sub-debt	Change
Equity cheque	£14,800k	£8,573k	(£6,227k)
Equity IRR	12.0%	13.9%	+195bps
Money multiple	1.81x	2.25x	+0.44x
Or: bid at an unchanged 12% IRR	£64,518k	£65,316k	+£798k

It is accretive because 9% is below the 12% hurdle and the interest is deductible, for an after-tax cost of about 6.75%. Use it either way — 195bps of return, or £0.8m more on price while still clearing 12%; in a competitive process the second is worth more. But total debt rises to £57.5m, the equity cushion falls to 13% of the price, and combined cover in 2017 is about 1.13x. It should not be committed in full without testing against the P90 case.

10. Risk

Holding the £64.5m price and committed debt constant and shocking the operating case:

Scenario	Equity IRR	Min DSCR	Outcome
Base case (P75)	12.0%	1.30x	On plan
P90 for 10 years, then P75	6.9%	1.21x	Lock-up risk
P90 every year	6.0%	1.09x	Covenant breach
Power price −5%	8.7%	1.10x	Covenant breach
Power price −10%	4.8%	0.89x	Payment default
Operating costs +10%	8.3%	1.02x	Covenant breach
Power price −10% and P90	(3.5%)	0.70x	Equity written off

The distribution is wide and asymmetric. A 10% price fall takes cover below 1.00x, while a 10% rise gives 17.7%. On the way down the lenders take the cash before the equity sees any of it.

Diligence priorities before bidding

- Technical due diligence on the load factor: P75 is the whole valuation, and four years of operating history should exist. If actual production has run below P75, the price comes down.
- Decommissioning: no provision is modelled. A £2–3m liability in 2027 is worth ~£0.5m today.
- Grid connection, O&M and PPA terms, and confirmation the subsidy is grandfathered to 2022.
- Share versus asset purchase, and the availability of capital allowances (conservatively assumed to follow book depreciation).

Appendix — Modelling judgements

Item	Treatment	Why
Base-case production	P75 (2,500 hours, 125 GWh p.a.)	Banks size on P75; P90 is the downside
Acquisition accounting	Share purchase, no step-up; PP&E £67.5m depreciating £4.5m p.a.	Carries the opening balance sheet forward; a step-up would not change the debt, since the DSCR test binds either way
Opening cash of £3.5m	Acquired and distributed in 2013	Inside the priced balance sheet; excluded from sizing
Working capital	Receivable and payable days derived from the opening balance sheet (36.9 and 35.0), held flat	Ties to the balance sheet rather than inventing a ratio; excluded from sizing
Upfront fee	Capitalised against the debt, amortised over 12 years, deductible within the cap	Amortised-cost treatment; keeps the balance sheet balanced without an extra asset line
Repayment profile	Sculpted to a constant DSCR	Maximises debt for a given cover level
Terminal value	Nil	Economic life ends 2027; no decommissioning cost modelled (see §10)
Electricity demand	+0.25% p.a. to 2015, +0.5% thereafter	CCGT keeps ample headroom above demand in every year, so the clearing plant is robust to the demand path

Model map: Assumptions (inputs) · Calcs (merit order, working capital, debt, tax) · Output (statements, valuation, sources & uses) · Analysis (headline results and supporting workings) · Sensitivities. Blue text is a hardcoded input, black a formula, green a cross-sheet link, yellow fill a key lever. Two cells are hardcoded by construction — the senior debt profile (Calcs) and the subordinated loan (Analysis D131) — because size, interest and tax are mutually dependent; each has a live sizing check beside it that must read nil.